		= 0.070 J
12	**D**	Option A is incorrect as Newton’s 1st law states the resultant force acting on the object at rest or moving at uniform velocity is zero. Option B is incorrect as Newton’s 3rd law deals with action-reaction pair of forces, not equilibrium of a system of forces. Option C is incorrect as in any system of force in equilibrium, the resultant force is zero. The forces can be different in magnitude. Option D is correct as the resultant force and the resultant torque are both zero in any system of forces in equilibrium.
13	**D**	Vector diagram of forces on the climber in equilibrium